

DataAces Recruitment Pattern

1. Recruitment Process at a Glance

Round	What students can expect	Primary focus	Priority
Round 1	Aptitude + Programming	Quantitative aptitude, logical reasoning, programming fundamentals	High
Round 2	Written Technical / Coding	Problem solving, algorithms, writing logic	Very High
Round 3	Technical + HR Interview	OOP, exception handling, projects, technologies, problem-solving approach	Very High

Important: This pattern is based on a recent candidate-reported DataAces Graduate Trainee campus process, not an official permanent syllabus. The company may modify the process by role, batch, or campus.

ROUND 1 — Aptitude + Programming

The reported first round included **Java, C, C++, Python**, along with **numerical and logical aptitude**.

A. Programming Fundamentals

Area	Important Topics	Priority
Programming Basics	Variables, data types, operators	High
Control Flow	if/else, loops, nested loops	High
Functions	Parameters, return values, scope	High
Arrays	Traversal, searching, basic manipulation	Very High
Strings	Character operations, manipulation	Very High
Recursion	Basic recursion/output tracing	Medium
Pointers	C/C++ basics	Medium
OOP Basics	Class, object, inheritance, polymorphism	High
Exception Handling	try/catch, throw/throws	High
Output Prediction	Trace code and determine output	Very High

Languages to be comfortable with

Students don't necessarily need equal depth in all four languages, but they should be able to **read and reason about code** in:

C → C++ → Java → Python

The reported DataAces first round specifically mentioned all four.



B. Numerical Aptitude

Focus on **problem-solving speed**, not advanced mathematics.

Must Prepare

1. Percentages
2. Profit & Loss
3. Ratio & Proportion
4. Averages
5. Time & Work
6. Time, Speed & Distance
7. Simple & Compound Interest
8. Number System
9. HCF & LCM
10. Permutation & Combination
11. Probability
12. Data Interpretation

Priority

Tier 1

- Percentages
- Ratio
- Average
- Profit & Loss
- Time & Work
- Time-Speed-Distance

Tier 2

- Probability
- P&C
- Number System
- SI/CI
- Data Interpretation

C. Logical Reasoning

High-priority areas

- Number series
- Coding-decoding
- Blood relations
- Directions
- Syllogisms

- Seating arrangements
- Puzzles
- Data sufficiency
- Statement & conclusion
- Pattern recognition

The goal should be:

Recognize the pattern → choose the approach → solve quickly.

ROUND 2 — Written Technical / Coding

This is where I would put the **largest preparation effort**.

The recent DataAces candidate report specifically describes this round as a **written exam where candidates have to write the logic of the problem**, with questions reported around **medium-to-hard LeetCode level**. The interviewer also emphasized problem-solving approach.

So students should **not prepare only by memorizing LeetCode solutions**.

They need to learn:

Problem → Observation → Approach → Algorithm → Code/Pseudocode → Complexity

ROUND 2 — DSA Priority

Tier 1 — Must Master

Topic	Priority
Arrays	★★★★★
Strings	★★★★★
Hashing	★★★★★
Two Pointers	★★★★★
Sliding Window	★★★★★
Sorting	★★★★★
Binary Search	★★★★★
Stack	★★★★☆
Queue	★★★★☆
Linked List	★★★★☆

These should form the **core DataAces coding preparation**.

Tier 2 — Strong Preparation

Topic	Priority
Recursion	★★★★☆
Backtracking	★★★★☆☆
Greedy	★★★★☆
Trees	★★★★☆
Heap / Priority Queue	★★★★☆
Intervals	★★★★☆
Prefix Sum	★★★★☆
Matrix	★★★★☆

Tier 3 — Advanced Preparation

Topic	Priority
Dynamic Programming	★★★★☆
Graphs	★★★★☆☆
Trie	★★☆☆☆☆
Union Find	★★☆☆☆☆
Advanced Graph Algorithms	★★☆☆☆☆

I would **not** start students with **DP and Graphs** just because the report says medium-hard.

The better progression is:

Arrays → Strings → Hashing → Two Pointer → Sliding Window → Binary Search → Stack → Linked List → Trees → Heap → Greedy → DP

That gives them the problem-solving patterns needed for the written round.

The Problem Patterns Students Must Recognize

This is more important than simply knowing 100 LeetCode problems.

Pattern 1 — Frequency / Hashing

Examples:

- Two Sum
- Valid Anagram
- First Unique Character
- Group Anagrams



Pattern 2 — Two Pointers

Examples:

- Two Sum II
- Valid Palindrome
- Container With Most Water
- 3Sum

Pattern 3 — Sliding Window

Examples:

- Longest Substring Without Repeating Characters
- Maximum Average Subarray
- Minimum Size Subarray Sum

Pattern 4 — Binary Search

Examples:

- Binary Search
- Search Insert Position
- Search in Rotated Sorted Array
- First/Last Position

Pattern 5 — Stack

Examples:

- Valid Parentheses
- Min Stack
- Daily Temperatures
- Next Greater Element

Pattern 6 — Intervals

Examples:

- Merge Intervals
- Insert Interval
- Non-overlapping Intervals

Pattern 7 — Greedy

Examples:

- Assign Cookies
- Jump Game
- Non-overlapping Intervals

Pattern 8 — DP

Examples:

- Climbing Stairs
- House Robber

- Coin Change

ROUND 3 — Technical + HR

The recent candidate report gives us a particularly useful signal here.

The reported areas were:

- OOP
- Exception Handling
- Technologies used in projects
- Resume
- Problem-solving approach

and the candidate reported that **most questions were asked from the resume**. ([Glassdoor](#))

Therefore, students should not treat this as a generic HR round.

Technical Areas

OOP — Very High Priority

Students must know:

- Class vs Object
- Constructor
- Encapsulation
- Inheritance
- Polymorphism
- Abstraction
- Method overloading
- Method overriding
- Interface vs abstract class
- Access modifiers
- this
- super
- Static members

More importantly:

"Give me a real-world example."

Students should be able to explain each concept with an example.

Exception Handling

Particularly important because it was explicitly mentioned in the recent candidate report.

Prepare:

- Exception vs Error
- try/catch
- finally
- throw
- throws
- Checked vs unchecked exceptions
- Multiple catch
- Custom exceptions
- Exception propagation

For Java students:

try

catch

finally

throw

throws

should be completely comfortable.

Projects — Very High Priority

Students should expect questions like:

Project Understanding

- What problem does your project solve?
- Why did you build it?
- What was your contribution?
- What technologies did you use?
- Why did you choose that technology?
- Explain the architecture.
- Explain your database.
- What was the biggest challenge?
- How did you solve it?
- What would you improve?
- What happens when 1,000 users access the system?
- What happens if the database fails?

The key rule:

If it is written on your resume, assume you can be questioned on it.

That is especially important given the recent DataAces candidate's report that most questions came from the resume.

DataAces Preparation Priority

If we convert the whole recruitment process into a student preparation priority:

Area	Priority	Suggested Weight in Preparation
DSA / Problem Solving	★★★★★	35%
Programming Fundamentals	★★★★★	20%
Aptitude	★★★★☆	15%
OOP + Exception Handling	★★★★☆	10%
Projects / Resume	★★★★★	10%
Other CS Fundamentals	★★★☆☆	5%
Communication / HR	★★★☆☆	5%

These percentages are **our preparation allocation**, not an official DataAces scoring scheme.